

Q1. In an expression $a \times 10^b$;

- (1) b is order of magnitude for $a \geq 5$ (2) b is order of magnitude for $a \leq 5$
 (3) a is order of magnitude for $b \leq 5$ (4) b is order of magnitude for $5 < a \leq 10$

Q2. Young's modulus is determined by the equation given by $Y = 49000 \frac{M}{l} \frac{\text{dyn}}{\text{cm}^2}$ where M is the mass and l is the extension of wire used in the experiment. Now error in Young modulus (Y) is estimated by taking data from $M - l$ plot in graph paper. The smallest scale divisions are 5 g and 0.02 cm along load axis and extension axis respectively. If the value of M and l are 500 g and 2 cm respectively then percentage error of Y is :

- (1) 0.5% (2) 2%
 (3) 0.02% (4) 0.2%

Q3. A clock has 75 cm, 60 cm long second hand and minute hand respectively. In 30 minutes duration the tip of second hand will travel x distance more than the tip of minute hand. The value of x in meter is nearly (Take

$\pi = 3.14$) :

- (1) 140.5 (2) 118.9
 (3) 139.4 (4) 220.0

Q4. A stationary particle breaks into two parts of masses m_A and m_B which move with velocities v_A and v_B respectively. The ratio of their kinetic energies ($K_B : K_A$) is :

- (1) $v_B : v_A$ (2) $m_B : m_A$
 (3) $m_B v_B : m_A v_A$ (4) 1 : 1

Q5. Three bodies A, B and C have equal kinetic energies and their masses are 400 g, 1.2 kg and 1.6 kg respectively.

The ratio of their linear momenta is :

- (1) $\sqrt{2}; \sqrt{3}; 1$ (2) $1 : \sqrt{3} : 2$
 (3) $1 : \sqrt{3} : \sqrt{2}$ (4) $\sqrt{3} : \sqrt{2} : 1$

Q6. A player caught a cricket ball of mass 150 g moving at a speed of 20 m/s. If the catching process is completed in 0.1 s, the magnitude of force exerted by the ball on the hand of the player is:

- (1) 3 N (2) 300 N
 (3) 150 N (4) 30 N

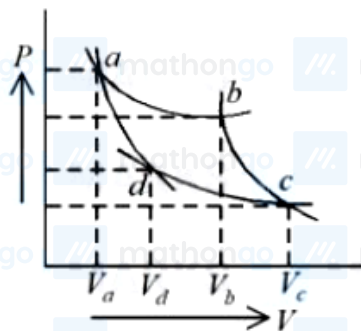
Q7. Two planets A and B having masses m_1 and m_2 move around the sun in circular orbits of r_1 and r_2 radii respectively. If angular momentum of A is L and that of B is $3L$, the ratio of time period $\left(\frac{T_A}{T_B}\right)$ is:

- (1) $\left(\frac{r_2}{r_1}\right)^{\frac{3}{2}}$ (2) $\frac{1}{27} \left(\frac{m_2}{m_1}\right)^3$
 (3) $27 \left(\frac{m_1}{m_2}\right)^3$ (4) $\left(\frac{r_1}{r_2}\right)^3$

Q8. Correct Bernoulli's equation is (symbols have their usual meaning) :

- (1) $P + mgh + \frac{1}{2}mv^2 = \text{constant}$ (2) $P + \rho gh + \frac{1}{2}\rho v^2 = \text{constant}$
 (3) $P + \rho gh + \rho v^2 = \text{constant}$ (4) $P + \frac{1}{2}\rho gh + \frac{1}{2}\rho v^2 = \text{constant}$

Q9. Two different adiabatic paths for the same gas intersect two isothermal curves as shown in P-V diagram. The



relation between the ratio $\frac{V_a}{V_d}$ and the ratio $\frac{V_b}{V_c}$ is:

(1) $\frac{V_a}{V_d} \neq \frac{V_b}{V_c}$

(2) $\frac{V_a}{V_d} = \frac{V_b}{V_c}$

(3) $\frac{V_a}{V_d} = \left(\frac{V_b}{V_c}\right)^{-1}$

(4) $\frac{V_a}{V_d} = \left(\frac{V_b}{V_c}\right)^2$

Q10. A mixture of one mole of monoatomic gas and one mole of a diatomic gas (rigid) are kept at room temperature (27°C). The ratio of specific heat of gases at constant volume respectively is:

(1) $\frac{7}{5}$

(2) $\frac{3}{5}$

(3) $\frac{5}{3}$

(4) $\frac{3}{2}$

Q11. Two charged conducting spheres of radii a and b are connected to each other by a conducting wire. The ratio of charges of the two spheres respectively is:

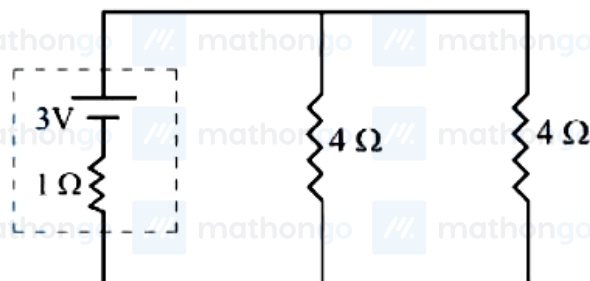
(1) $\frac{a}{b} v$

(2) $\sqrt{ab}$

(3) $\frac{b}{a}$

(4) ab

Q12. In the given circuit, the terminal potential difference of the cell is :



(1) 2 V

(2) 3 V

(3) 4 V

(4) 1.5 V

Q13. Paramagnetic substances: A. align themselves along the directions of external magnetic field. B. attract strongly towards external magnetic field. C. has susceptibility little more than zero. D. move from a region of strong magnetic field to weak magnetic field. Choose the most appropriate answer from the options given below:

(1) A, B, C Only

(2) A, B, C, D

(3) A, C Only

(4) B, D Only

Q14. A LCR circuit is at resonance for a capacitor C , inductance L and resistance R . Now the value of resistance is halved keeping all other parameters same. The current amplitude at resonance will be now:

- (1) Zero
(2) same
(3) halved
(4) double

Q15. Critical angle of incidence for a pair of optical media is 45° . The refractive indices of first and second media are in the ratio:

- (1) $1 : \sqrt{2}$
(2) $\sqrt{2} : 1$
(3) $2 : 1$
(4) $1 : 2$

Q16. A proton and an electron are associated with same de-Broglie wavelength. The ratio of their kinetic energies is:

(Assume $h = 6.63 \times 10^{-34}$ J s, $m_e = 9.0 \times 10^{-31}$ kg and $m_p = 1836$ times m_e)

- (1) $1 : \sqrt{1836}$
(2) $1 : \frac{1}{1836}$
(3) $1 : \frac{1}{\sqrt{1836}}$
(4) $1 : 1836$

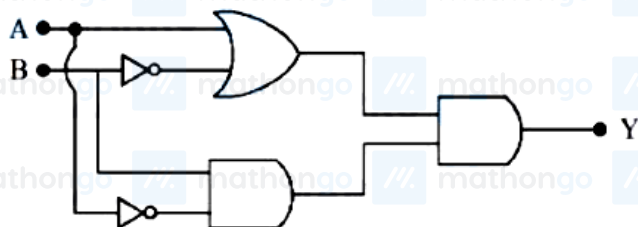
Q17. Average force exerted on a non-reflecting surface at normal incidence is 2.4×10^{-4} N. If 360 W/cm^2 is the light energy flux during span of 1 hour 30 minutes, Then the area of the surface is:

- (1) 0.2 m^2
(2) 20 m^2
(3) 0.1 m^2
(4) 0.02 m^2

Q18. Binding energy of a certain nucleus is 18×10^8 J. How much is the difference between total mass of all the nucleons and nuclear mass of the given nucleus:

- (1) $10 \mu\text{g}$
(2) $20 \mu\text{g}$
(3) $0.2 \mu\text{g}$
(4) $2 \mu\text{g}$

Q19. The output Y of following circuit for given inputs is :

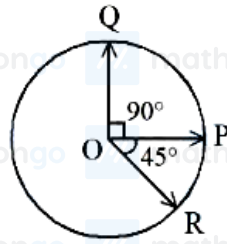


- (1) $A \cdot B(A + B)$
(2) 0
(3) $\bar{A} \cdot B$
(4) $A \cdot B$

Q20. The diameter of a sphere is measured using a vernier caliper whose 9 divisions of main scale are equal to 10 divisions of vernier scale. The shortest division on the main scale is equal to 1 mm. The main scale reading is 2 cm and second division of vernier scale coincides with a division on main scale. If mass of the sphere is 8.635 g, the density of the sphere is:

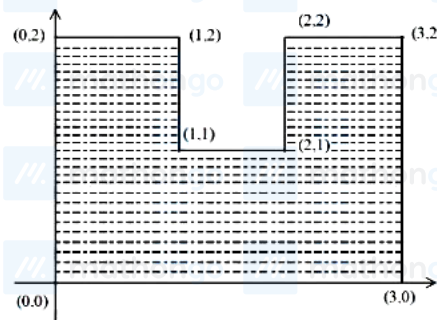
- (1) 2.0 g/cm^3
(2) 1.7 g/cm^3
(3) 2.2 g/cm^3
(4) 2.5 g/cm^3

- Q21. Three vectors $\vec{OP}$, $\vec{OQ}$ and $\vec{OR}$ each of magnitude A are acting as shown in figure. The resultant of the three



vectors is $A\sqrt{x}$. The value of x is _____.

- Q22. A uniform thin metal plate of mass 10 kg with dimensions is shown. The ratio of x and y coordinates of center



of mass of plate in $\frac{n}{9}$. The value of n is _____

- Q23. A liquid column of height 0.04 cm balances excess pressure of a soap bubble of certain radius. If density of liquid is $8 \times 10^3 \text{ kg m}^{-3}$ and surface tension of soap solution is 0.28 Nm^{-1} , then diameter of the soap bubble is _____ cm. (if $g = 10 \text{ m s}^{-2}$)

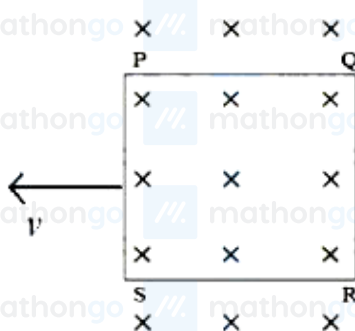
- Q24. A closed and an open organ pipe have same lengths. If the ratio of frequencies of their seventh overtones is $\left(\frac{a-1}{a}\right)$ then the value of a is _____

- Q25. An electric field, $\vec{E} = \frac{2\hat{i} + 6\hat{j} + 8\hat{k}}{\sqrt{6}}$ passes through the surface of 4 m^2 area having unit vector $\hat{n} = \left(\frac{2\hat{i} + \hat{j} + \hat{k}}{\sqrt{6}}\right)$. The electric flux for that surface is _____ Vm.

- Q26. Resistance of a wire at 0°C , 100°C and $t^\circ\text{C}$ is found to be 10Ω , 10.2Ω and 10.95Ω respectively. The temperature t in Kelvin scale is _____

- Q27. An electron with kinetic energy 5eV enters a region of uniform magnetic field of $3 \mu\text{T}$ perpendicular to its direction. An electric field E is applied perpendicular to the direction of velocity and magnetic field. The value of E , so that electron moves along the same path, is _____ NC^{-1} . (Given, mass of electron = $9 \times 10^{-31} \text{ kg}$, electric charge = $1.6 \times 10^{-19} \text{ C}$)

- Q28. A square loop PQRS having 10 turns, area $3.6 \times 10^{-3} \text{ m}^2$ and resistance 100Ω is slowly and uniformly being pulled out of a uniform magnetic field of magnitude $B = 0.5 \text{ T}$ as shown. Work done in pulling the loop out of



the field in 1.0 s is $\times 10^{-6}$ J.

Q29. A parallel beam of monochromatic light of wavelength 600 nm passes through single slit of 0.4 mm width.

Angular divergence corresponding to second order minima would be $\times 10^{-3}$ rad.

Q30. In an alpha particle scattering experiment distance of closest approach for the α particle is 4.5×10^{-14} m. If

target nucleus has atomic number 80, then maximum velocity of α - particle is $\times 10^5$ m/s

approximately. $\left(\frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ SI unit, mass of } \alpha \text{ particle} = 6.72 \times 10^{-27} \text{ kg}\right)$

Q31. Combustion of glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) produces CO_2 and water. The amount of oxygen (in g) required for the complete combustion of 900 g of glucose is : [Molar mass of glucose in $\text{gmol}^{-1} = 180$]

(1) 480

(2) 800

(3) 960

(4) 32

Q32.

Match List I with List II

	List - I		List - II
	(Elements)		(Properties in their respective groups)
A.	Cl, S	I.	Elements with highest electronegativity
B.	Ge, As	II.	Elements with largest atomic size
C.	Fr, Ra	III.	Elements which show properties of both metals and non-metal
D.	F, O	IV.	Elements with highest negative electron gain enthalpy

Choose

the correct answer from the options given below:

(1) A-II, B-I, C-IV, D-III

(2) A-III, B-II, C-I, D-IV

(3) A-IV, B-III, C-II, D-I

(4) A-II, B-III, C-IV, D-I

Q33.

Match List I with List II

	List-I		List-II
	(Molecule)		(Shape)
A.	NH_3	I.	Square pyramid
B.	BrF_5	II.	Tetrahedral
C.	PCl_5	III.	Trigonal pyramidal
D.	CH_4	IV.	Trigonal bipyramidal

Choose the correct answer

from the options given below:

(1) A-II, B-IV, C-I, D-III

(2) A-III, B-I, C-IV, D-II

(3) A-IV, B-III, C-I, D-II

(4) A-III, B-IV, C-I, D-II

Q34.

Match List I with List II

	List-I (Molecule)		List-II (Shape)
A.	$\text{Fe}_4[\text{Fe}(\text{CN})_6]_3 \cdot x\text{H}_2\text{O}$	I.	Violet
B.	$[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$	II.	Blood Red A
C.	$[\text{Fe}(\text{SCN})]^{2+}$	III.	Prussian Blue
D.	$(\text{NH}_4)_3\text{PO}_4 \cdot 12\text{MoO}_3$	IV.	Yellow

Choose the correct answer

from the options given below:

(1) A-III, B-I, C-II, D-IV

(2) A-I, B-II, C-III, D-IV

(3) A-IV, B-I, C-II, D-III

(4) A-II, B-III, C-IV, D-I

Q35. Given below are two statements: Statement I: $\text{N}(\text{CH}_3)_3$ and $\text{P}(\text{CH}_3)_3$ can act as ligands to form transition metal complexes. Statement II: As N and P are from same group, the nature of bonding of $\text{N}(\text{CH}_3)_3$ and $\text{P}(\text{CH}_3)_3$ is always same with transition metals. In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Statement I is correct but Statement II is incorrect.

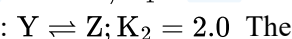
(2) Statement I is incorrect but Statement II is correct.

(3) Both Statement I and Statement II are correct.

(4) Both Statement I and Statement II are incorrect.

Q36.

For the given hypothetical reactions, the equilibrium constants are as follows : $\text{Y} \rightleftharpoons \text{Z}; K_2 = 2.0$ The equilibrium constant for the reaction $\text{X} \rightleftharpoons \text{W}$ is



(1) 6.0

(2) 12.0

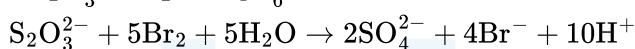
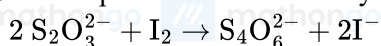
(3) 7.0

(4) 8.0

Q37. Among the following halogens F_2 , Cl_2 , Br_2 and I_2 Which can undergo disproportionation reactions?

(1) F_2 , Cl_2 and Br_2 (2) F_2 and Cl_2 (3) Only I_2 (4) Cl_2 , Br_2 and I_2

Q38. Thiosulphate reacts differently with iodine and bromine in the reactions given below:



behaviour of thiosulphate?

Which of the following statement justifies the above dual

(1) Bromine is a stronger oxidant than iodine

(2) Thiosulphate undergoes oxidation by bromine and reduction by iodine in these reaction

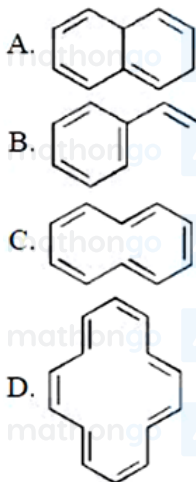
(3) Bromine is a weaker oxidant than iodine

(4) Bromine undergoes oxidation and iodine undergoes reduction in these reactions

Q39. Give below are two statements: One is labelled as Assertion A and the other is labelled as Reason R: Assertion A: The stability order of +1 oxidation state of Ga, In and Tl is $\text{Ga} < \text{In} < \text{Tl}$. Reason R: The inert pair effect stabilizes the lower oxidation state down the group. In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false. (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A. (4) Both A and R are true but R is NOT the correct explanation of A.

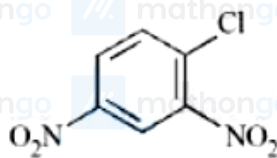
Q40.



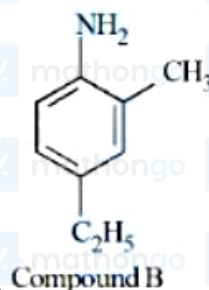
Which of the following are aromatic?

- (1) A and C only (2) B and D only
 (3) C and D only (4) A and B only

Q41.



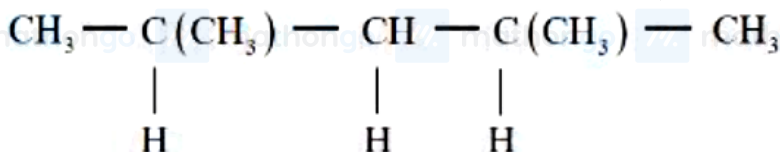
Given below are two statements: Statements I:



IUPAC name of Compound A is 4-chloro-1,3-dinitrobenzene. Statements II:

IUPAC name of Compound B is 4-ethyl-2-methylaniline. In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both Statement I and Statement II are incorrect. (2) Both Statement I and Statement II are correct.
 (3) Statement I is correct but Statement II is incorrect. (4) Statement I is incorrect but Statement II is correct.

Q42. In the given compound, the number of 2° carbon atom /s is _____.

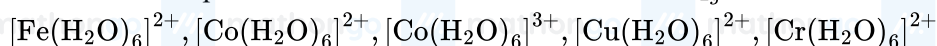
- (1) Four (2) Two
 (3) One (4) Three

Q43. Iron (III) catalyses the reaction between iodide and persulphate ions, in which A. Fe^{3+} oxidises the iodide ion
 B. Fe^{3+} oxidises the persulphate ion C. Fe^{2+} reduces the iodide ion D. Fe^{2+} reduces the persulphate ion

Choose the most appropriate answer from the options given below:

- (1) B only (2) A only
 (3) B and C only (4) A and D only

Q44. Number of Complexes with even number of electrons in t_{2g} orbitals is -



- (1) 2 (2) 3
 (3) 1 (4) 5

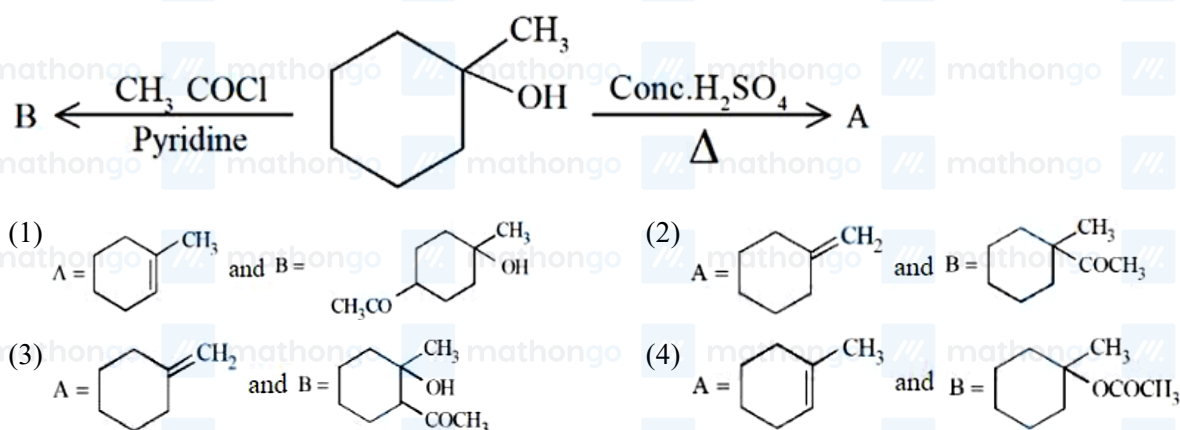
Q45. An octahedral complex with the formula $\text{CoCl}_3 \cdot n\text{NH}_3$ upon reaction with excess of AgNO_3 solution gives 2 moles of AgCl . Consider the oxidation state of Co in the complex is 'x'. The value of "x + n" is _____

- (1) 6 (2) 8
 (3) 3 (4) 5

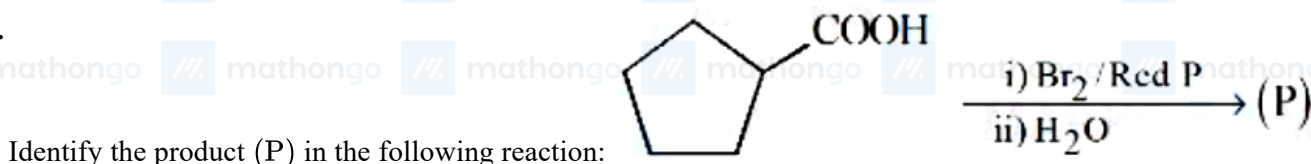
Q46. Which among the following compounds will undergo fastest $\text{S}_{\text{N}}2$ reaction.

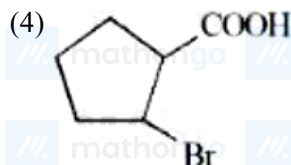
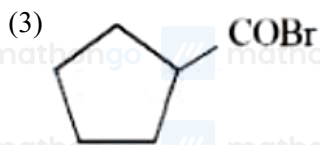


Q47. Identify the major products A and B respectively in the following set of reactions.



Q48.





Q49.

Match List I with List II

	List-I		List-II
	(Name of the test)		(Reaction sequence involved) [M is metal]
A.	Borax bead test	I.	$\text{MCO}_3 \rightarrow \text{MO} \xrightarrow{\text{Co(NO}_3)_2} \text{CoO} \cdot \text{MO}$
B.	Charcoal cavity test	II.	$\text{MCO}_3 \rightarrow \text{MCl}_2 \rightarrow \text{M}^{2+}$
C.	Cobalt nitrate test	III.	$\text{MSO}_4 \xrightarrow[\Delta]{\text{Na}_2\text{B}_4\text{O}_7} \text{M(BO}_2)_2 \rightarrow \text{MBO}_2 \rightarrow \text{M}$
D.	Flame test	IV.	$\text{MSO}_4 \xrightarrow[\Delta]{\text{Na}_2\text{CO}_3} \text{MCO}_3 \rightarrow \text{MO} \rightarrow \text{M}$

Choose the correct answer from the options given below:

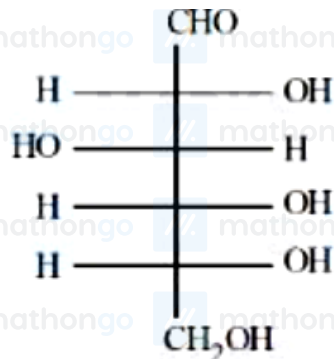
(1) A-III, B-II, C-IV, D-I

(2) A-III, B-IV, C-I, D-II

(3) A-III, B-I, C-II, D-IV

(4) A-III, B-I, C-IV, D-II

Q50.



The incorrect statement regarding the given structure is

(1) can be oxidized to a dicarboxylic acid with Br_2

(2) will coexist in equilibrium with 2 other cyclic

water

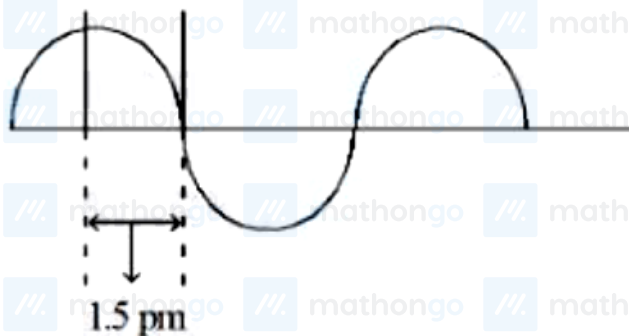
structure

(3) despite the presence of -CHO does not give

(4) has 4 asymmetric carbon atom

Schiff's test

Q51.



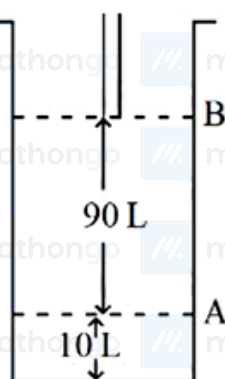
A hypothetical electromagnetic wave is shown below.

The frequency of the wave is $x \times 10^{19}$ Hz. $x =$ _____ (nearest integer)

Q52. Number of molecules from the following which are exceptions to octet rule is _____

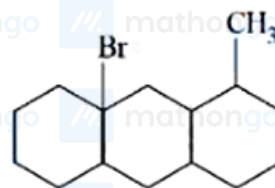
$\text{CO}_2, \text{NO}_2, \text{H}_2\text{SO}_4, \text{BF}_3, \text{CH}_4, \text{SiF}_4, \text{ClO}_2, \text{PCl}_5, \text{BeF}_2, \text{C}_2\text{H}_6, \text{CHCl}_3, \text{CBr}_4$

Q53.



Consider the figure provided. 1 mol of an ideal gas is kept in a cylinder, fitted with a piston, at the position A, at 18°C . If the piston is moved to position B, keeping the temperature unchanged, then ' x ' L atm work is done in this reversible process. $x =$ _____ L atm. (nearest integer) [Given : Absolute temperature $= ^\circ\text{C} + 273.15$, $R = 0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$]

Q54.



The number of optical isomers in following compound is: _____

Q55. A solution containing 10 g of an electrolyte AB_2 in 100 g of water boils at 100.52°C . The degree of ionization of the electrolyte (α) is _____ $\times 10^{-1}$. (nearest integer) [Given : Molar mass of $\text{AB}_2 = 200 \text{ g mol}^{-1}$, K_b (molal boiling point elevation const. of water) $= 0.52 \text{ K kg mol}^{-1}$, boiling point of water $= 100^\circ\text{C}$; AB_2 ionises as $\text{AB}_2 \rightarrow \text{A}^{2+} + 2 \text{B}^-$]

Q56. Consider the following reaction

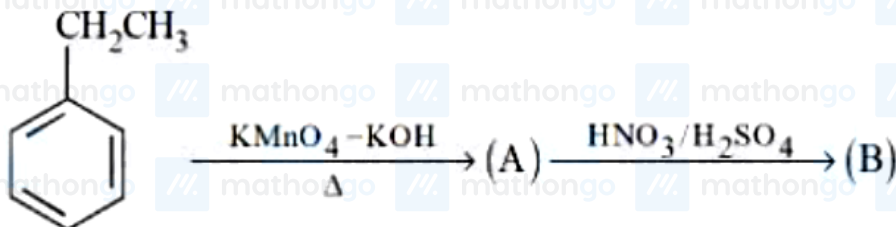


The time taken for A to become $1/4^{\text{th}}$ of its initial concentration is twice the time taken to become $1/2$ of the same. Also, when the change of concentration of B is plotted against time, the resulting graph gives a straight

line with a negative slope and a positive intercept on the concentration axis. The overall order of the reaction is _____

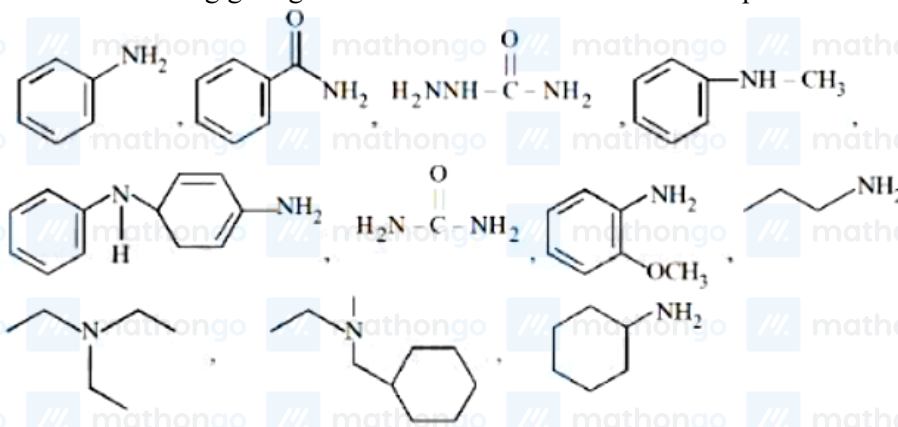
Q57. The 'spin only' magnetic moment value of MO_4^{2-} is _____ BM. (Where M is a metal having least metallic radii. among Sc, Ti, V, Cr, Mn and Zn). (Given atomic number: Sc = 21, Ti = 22, V = 23, Cr = 24, Mn = 25 and Zn = 30)

Q58. Major product *B* of the following reaction has _____ π -bond.



Q59. If 279 g of aniline is reacted with one equivalent of benzenediazonium chloride, the maximum amount of aniline yellow formed will be _____ g. (nearest integer) (consider complete conversion).

Q60. Number of amine compounds from the following giving solids which are soluble in NaOH upon reaction with



Hinsberg's reagent is _____

Q61. The sum of all the solutions of the equation $(8)^{2x} - 16 \cdot (8)^x + 48 = 0$ is :

- (1) $1 + \log_8(6)$ (2) $1 + \log_6(8)$
(3) $\log_8(6)$ (4) $\log_8(4)$

Q62. Let z be a complex number such that $|z + 2| = 1$ and $\text{Im}\left(\frac{z+1}{z+2}\right) = \frac{1}{5}$. Then the value of $|\text{Re}(z + 2)|$ is

- (1) $\frac{2\sqrt{6}}{5}$ (2) $\frac{24}{5}$
(3) $\frac{1+\sqrt{6}}{5}$ (4) $\frac{\sqrt{6}}{5}$

Q63. If the set $R = \{(a, b) : a + 5b = 42, a, b \in \mathbb{N}\}$ has m elements and $\sum_{n=1}^m (1 - i^n) = x + iy$, where $i = \sqrt{-1}$, then the value of $m + x + y$ is

- (1) 12 (2) 4
(3) 8 (4) 5

Q64. If $\sin x = -\frac{3}{5}$, where $\pi < x < \frac{3\pi}{2}$, then $80(\tan^2 x - \cos x)$ is equal to

- (1) 108 (2) 109
(3) 18 (4) 19

Q65. The equations of two sides AB and AC of a triangle ABC are $4x + y = 14$ and $3x - 2y = 5$, respectively. The point $(2, -\frac{4}{3})$ divides the third side BC internally in the ratio 2 : 1. the equation of the side BC is

- (1) $x + 3y + 2 = 0$ (2) $x - 6y - 10 = 0$
 (3) $x - 3y - 6 = 0$ (4) $x + 6y + 6 = 0$

Q66. Let the circles $C_1 : (x - \alpha)^2 + (y - \beta)^2 = r_1^2$ and $C_2 : (x - 8)^2 + (y - \frac{15}{2})^2 = r_2^2$ touch each other externally at the point (6, 6). If the point (6, 6) divides the line segment joining the centres of the circles C_1 and C_2 internally in the ratio 2 : 1, then $(\alpha + \beta) + 4(r_1^2 + r_2^2)$ equals

- (1) 125 (2) 130
 (3) 110 (4) 145

Q67. Let $H : \frac{-x^2}{a^2} + \frac{y^2}{b^2} = 1$ be the hyperbola, whose eccentricity is $\sqrt{3}$ and the length of the latus rectum is $4\sqrt{3}$. Suppose the point $(\alpha, 6)$, $\alpha > 0$ lies on H . If β is the product of the focal distances of the point $(\alpha, 6)$, then $\alpha^2 + \beta$ is equal to

- (1) 172 (2) 171
 (3) 169 (4) 170

Q68. Let $A = \begin{bmatrix} 2 & a & 0 \\ 1 & 3 & 1 \\ 0 & 5 & b \end{bmatrix}$. If $A^3 = 4A^2 - A - 21I$, where I is the identity matrix of order 3×3 , then $2a + 3b$ is

equal to

- (1) -9 (2) -13
 (3) -10 (4) -12

Q69. Let $[t]$ be the greatest integer less than or equal to t . Let A be the set of all prime factors of 2310 and

$f : A \rightarrow \mathbb{Z}$ be the function $f(x) = \left\lceil \log_2 \left(x^2 + \left\lceil \frac{x^3}{5} \right\rceil \right) \right\rceil$. The number of one-to-one functions from A to the range of f is

- (1) 25 (2) 24
 (3) 20 (4) 120

Q70. For the function $f(x) = (\cos x) - x + 1$, $x \in \mathbb{R}$, between the following two statements (S1) $f(x) = 0$ for only one value of x in $[0, \pi]$. (S2) $f(x)$ is decreasing in $[0, \frac{\pi}{2}]$ and increasing in $[\frac{\pi}{2}, \pi]$.

- (1) Both (S1) and (S2) are correct. (2) Both (S1) and (S2) are incorrect.
 (3) Only (S2) is correct. (4) Only (S1) is correct.

Q71. Let $f(x) = 4 \cos^3 x + 3\sqrt{3} \cos^2 x - 10$. The number of points of local maxima of f in interval $(0, 2\pi)$ is

- (1) 3 (2) 4
 (3) 1 (4) 2

Q72. The number of critical points of the function $f(x) = (x - 2)^{2/3}(2x + 1)$ is

- (1) 1 (2) 2
 (3) 0 (4) 3

Q73. Let $I(x) = \int \frac{6}{\sin^2 x (1 - \cot x)^2} dx$. If $I(0) = 3$, then $I(\frac{\pi}{12})$ is equal to

(1) $2\sqrt{3}$

(3) $3\sqrt{3}$

(2) $\sqrt{3}$

(4) $6\sqrt{3}$

Q74. The value of $k \in \mathbb{N}$ for which the integral $I_n = \int_0^1 (1-x^k)^n dx$, $n \in \mathbb{N}$, satisfies $147I_{20} = 148I_{21}$ is

(1) 14

(2) 8

(3) 10

(4) 7

Q75. Let $f(x)$ be a positive function such that the area bounded by $y = f(x)$, $y = 0$ from $x = 0$ to $x = a > 0$ is $e^{-a} + 4a^2 + a - 1$. Then the differential equation, whose general solution is $y = c_1 f(x) + c_2$, where c_1 and c_2 are arbitrary constants, is

(1) $(8e^x - 1) \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

(2) $(8e^x - 1) \frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

(3) $(8e^x + 1) \frac{d^2y}{dx^2} - \frac{dy}{dx} = 0$

(4) $(8e^x + 1) \frac{d^2y}{dx^2} + \frac{dy}{dx} = 0$

Q76. Let $y = y(x)$ be the solution of the differential equation

$$(1+y^2)e^{\tan x} dx + \cos^2 x (1+e^{2\tan x}) dy = 0, y(0) = 1. \text{ Then } y\left(\frac{\pi}{4}\right) \text{ is equal to}$$

(1) $\frac{2}{e}$

(2) $\frac{2}{e^2}$

(3) $\frac{1}{e}$

(4) $\frac{1}{e^2}$

Q77. The set of all α , for which the vectors $\vec{a} = \alpha t \hat{i} + 6\hat{j} - 3\hat{k}$ and $\vec{b} = t\hat{i} - 2\hat{j} - 2\alpha t \hat{k}$ are inclined at an obtuse angle for all $t \in \mathbb{R}$, is

(1) $\left(-\frac{4}{3}, 1\right)$

(2) $[0, 1)$

(3) $\left(-\frac{4}{3}, 0\right]$

(4) $(-2, 0]$

Q78. If the shortest distance between the lines $L_1 : \vec{r} = (2 + \lambda)\hat{i} + (1 - 3\lambda)\hat{j} + (3 + 4\lambda)\hat{k}$, $\lambda \in \mathbb{R}$ is $\frac{m}{\sqrt{n}}$
 $L_2 : \vec{r} = 2(1 + \mu)\hat{i} + 3(1 + \mu)\hat{j} + (5 + \mu)\hat{k}$, $\mu \in \mathbb{R}$

, where $\gcd(m, n) = 1$, then the value of $m + n$ equals

(1) 390

(2) 384

(3) 377

(4) 387

Q79. Let $P(x, y, z)$ be a point in the first octant, whose projection in the xy -plane is the point Q . Let $OP = \gamma$; the angle between OQ and the positive x -axis be θ ; and the angle between OP and the positive z -axis be ϕ , where O is the origin. Then the distance of P from the x -axis is

(1) $\gamma \sqrt{1 - \sin^2 \phi \cos^2 \theta}$

(2) $\gamma \sqrt{1 - \sin^2 \theta \cos^2 \phi}$

(3) $\gamma \sqrt{1 + \cos^2 \phi \sin^2 \theta}$

(4) $\gamma \sqrt{1 + \cos^2 \theta \sin^2 \phi}$

Q80. Let the sum of two positive integers be 24. If the probability, that their product is not less than $\frac{3}{4}$ times their greatest possible product, is $\frac{m}{n}$, where $\gcd(m, n) = 1$, then $n - m$ equals

(1) 10

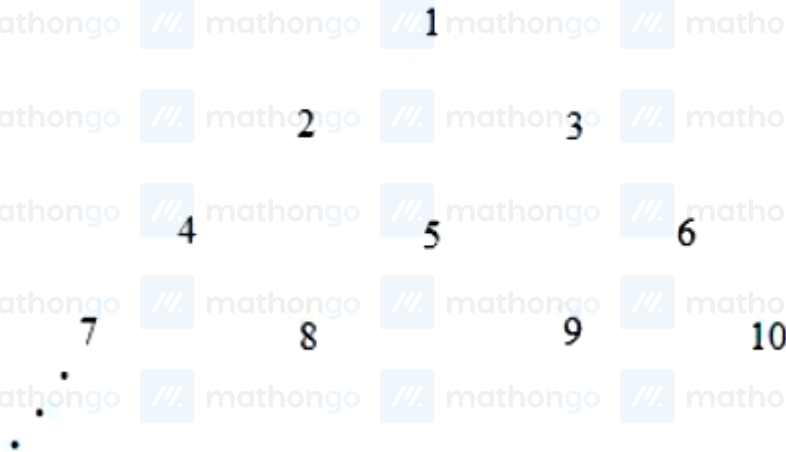
(2) 9

(3) 11

(4) 8

Q81. The number of 3-digit numbers, formed using the digits 2, 3, 4, 5 and 7, when the repetition of digits is not allowed, and which are not divisible by 3, is equal to _____

Q82. Let the positive integers be written in the form :



If the k^{th} row contains exactly k numbers for every natural number k , then the row in which the number 5310 will be, is _____

Q83. Let $\alpha = \sum_{r=0}^n (4r^2 + 2r + 1)^n C_r$ and $\beta = \left(\sum_{r=0}^n \frac{n C_r}{r+1} \right) + \frac{1}{n+1}$. If $140 < \frac{2\alpha}{\beta} < 281$, then the value of n is _____

Q84. If the orthocentre of the triangle formed by the lines $2x + 3y - 1 = 0$, $x + 2y - 1 = 0$ and $ax + by - 1 = 0$, is the centroid of another triangle, whose circumcentre and orthocentre respectively are $(3, 4)$ and $(-6, -8)$, then the value of $|a - b|$ is _____

Q85. The value of $\lim_{x \rightarrow 0} 2 \left(\frac{1 - \cos x \sqrt{\cos 2x} \sqrt[3]{\cos 3x} \dots \sqrt[10]{\cos 10x}}{x^2} \right)$ is _____

Q86. Let $A = \begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix}$. If the sum of the diagonal elements of A^{13} is 3^n , then n is equal to _____

Q87. If the range of $f(\theta) = \frac{\sin^4 \theta + 3 \cos^2 \theta}{\sin^4 \theta + \cos^2 \theta}$, $\theta \in \mathbb{R}$ is $[\alpha, \beta]$, then the sum of the infinite G.P., whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to _____

Q88. Let the area of the region enclosed by the curve $y = \min\{\sin x, \cos x\}$ and the x axis between $x = -\pi$ to $x = \pi$ be A . Then A^2 is equal to _____

Q89. Let $\vec{a} = 9\hat{i} - 13\hat{j} + 25\hat{k}$, $\vec{b} = 3\hat{i} + 7\hat{j} - 13\hat{k}$ and $\vec{c} = 17\hat{i} - 2\hat{j} + \hat{k}$ be three given vectors. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} = (\vec{b} + \vec{c}) \times \vec{a}$ and $\vec{r} \cdot (\vec{b} - \vec{c}) = 0$, then $\frac{|593\vec{r} + 67\vec{a}|^2}{(593)^2}$ is equal to _____

Q90. Three balls are drawn at random from a bag containing 5 blue and 4 yellow balls. Let the random variables X and Y respectively denote the number of blue and yellow balls. If $\bar{X}$ and $\bar{Y}$ are the means of X and Y respectively, then $7\bar{X} + 4\bar{Y}$ is equal to _____